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Case 8 — Ms. Chloe Davies | 27 F · Functional Neurological Disorder (FND) — Motor Subtype

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

- Left leg weakness — inconsistent, attention-modulated
- Left arm postural/action tremor — frequency entrainable
- Abnormal gait — left foot drag pattern
- Possible secondary deconditioning from activity avoidance
- Altered body-schema / limb awareness for left side
- Difficulty walking safely and efficiently in the community
- Left arm tremor limits fine motor tasks and carrying objects
- Reduced independence in ADLs and community participation
- Potential impact on employment and social roles
- Fear-avoidance of movement maintaining disability cycle

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

• In FND, this system becomes dysregulated — the brain generates an overpowering, fixed prior that the limb cannot move normally, and this prediction overrides normal voluntary motor commands.
• Functionally, this manifests as symptoms that are real, involuntary, and neurologically generated, but not driven by structural damage to any specific pathway.
• This is consistent with the key clinical feature: symptoms fluctuate with attention — they improve when Chloe is distracted and worsen when she focuses on them or feels observed.
• The precipitating psychological stress likely sensitised this prefrontal-limbic-motor circuit, functioning as a "trigger" event for the disorder.

■ DIFFERENTIAL DIAGNOSES

1. Confirmed by positive clinical signs:
- Confirmed by positive clinical signs:

2. Supporting features initially:
- Supporting features initially:

3. Supporting features initially:
- Supporting features initially:

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Hoover Sign
MEASURES Positive FND motor sign — tests whether lower limb weakness is voluntary (intentional motor drive absent) or associated with intact automatic motor output HOW TO PERFORM Patient lies supine. Explain you are going to test the strength in both legs. Place one hand under th

Tremor Entrainment Test
MEASURES Distinguishes functional tremor (frequency entrains to external rhythm) from organic tremors (fixed frequency driven by a subcortical oscillator) HOW TO PERFORM Ask Chloe to hold the affected left arm outstretched so the tremor is visible. Observe and note the resting tr

Gait Observation: Inconsistency Assessment
MEASURES Documents inconsistency in the gait impairment (a positive FND feature) and baselines the gait pattern as a treatment target HOW TO PERFORM Observe Chloe walking from one end of the room to the other with no instruction other than to walk normally — document foot clearan

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4 (2 sessions/week)
- Neurophysio Rationale for FND
- FND physiotherapy is a
- bottom-up motor retraining approach
- Exercise 1 — Rhythmic Auditory-Cued Walking (Automatic Gait Initiation)
- ■ Regression

■ Long-Term Plan — Weeks 4–12+
- neurosymptoms.org
- (developed by leading FND neurologist Professor Jon Stone, University of Edinburgh) and the
- patient charity network. Regular physiotherapy review every 6–8 weeks as Chloe develops self-management skills, with the explicit goal of reducing dependence on formal therapy as recovery progresses.

- ■ Examiner Demonstration — Techniques Performable in the Room
- Tremor Entrainment Live Demonstration: